

A rational solution to an agreement-interpretation puzzle

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1 Introduction

- ¹FRACTION PARTITIVES (FPs) which embed a plural complement are commonly understood to denote a subset of their complement whose cardinality is that of the complement multiplied by the fraction (see, e.g., Wagiel 2021; Ionin et al. 2006; Benbaji-Elhadad and Wehbe 2024)
- (1) A third of the apples $\approx$ 2 out of 6 apples, 3 out of 9 apples, etc.
- This is a fairly comprehensive picture for languages like American English
- However, the FPs of some languages exhibit an additional reading which often goes unnoticed in the literature (save for some work which I will discuss in depth a bit later)
- Consider a hexagonal studio apartment with a floor plan like that in *fig. 1*

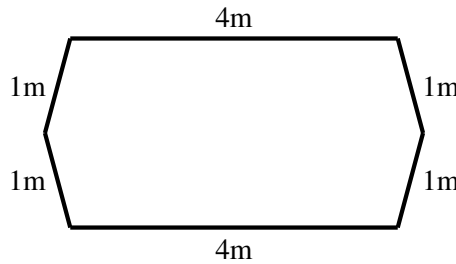


Figure 1: Floor plan of your studio apartment (not to scale)

- Now imagine that this apartment has developed a mold problem; the blight has covered a third of the walls
- If you told your Italian-speaking landlord (2) over the phone, the situation is compatible with two scenarios—one where two out of the six walls are covered in mold (*fig. 2a*), and one where a large wall whose surface area is a third of that of the combined walls is covered in mold (*fig. 2b*)
- I call the first a COUNT reading and the second a MEASURE reading²

(2) la muffa ha coperto un terz-o del-le paret-i
 DET.F.SG mold has.3SG covered.3SG one third-M.SG of-DET.F.PL wall-F.PL
 ‘The mold has covered a third of the walls’ [Italian]

¹This is my longest-running project to date, so there are many people to thank. This research has benefited immensely from conversations with Nina Haslinger, Danny Fox, Martin Hackl, Sabine Iatridou, David Pesetsky, Kai von Fintel, Marcin Wągiel, Yasu Sudo, Enrico Flor, Paul Meisenbichler, Yiannis Katochoritis, Peter Grishin, David Adger, Gereon Müller, Adam Przepiorkowski, and Shigeru Miyagawa. I would also like to thank audiences at the CreteLing Conference 2024, the 2024 *Puzzles of Agreement* workshop, MIT LFRG, and the Spring 2025 MIT Workshop for their insight and feedback. For their judgments and patience, I am indebted to Lorenzo Pinton, Enrico Flor, Yiannis Katochoritis, Sabine Iatridou, Vana Tsakali, Adèle Mortier, Genia Lukin, Vlad Orlov, Paul Meisenbichler, Nina Haslinger, and Martin Hackl. Errors are solely my responsibility. If you have a comment, thought, or judgment which you were unable to share in-person, please write to me at jcrobert [at] mit [dot] edu.

²My use of *unambiguously* in *fig. 2a, 2b* should be clear; if the apartment layout was s.t. every member of the wall plurality was congruent (same size and shape), then there would be substantial overlap between the count and measure reading.

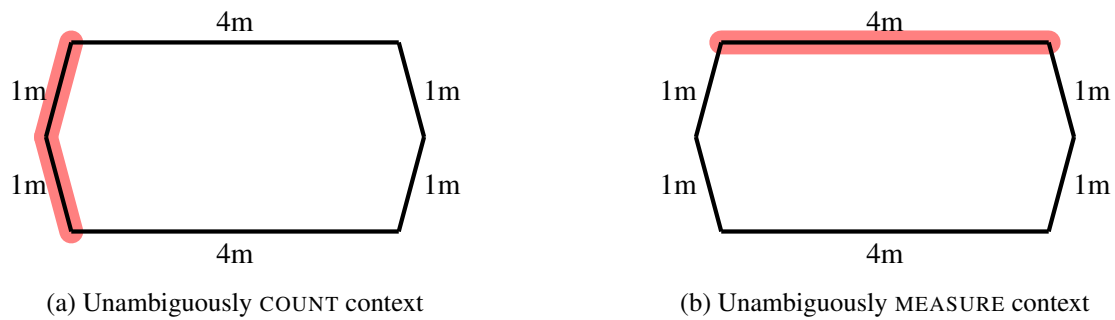


Figure 2: Two possible scenarios for mold coverage

- **Where it gets interesting:** in a position of agreement, you can disambiguate between the two readings

- Copy the features of the denominator³ $\Rightarrow$ only the measure interpretation
- Copy the features of the complement $\Rightarrow$ only the count interpretation

- (3) ‘One third of the walls are covered by mold’ [Italian]

- a. (esattamente) un **terz-o** del-le paret-i è coperto di muffa
 (exactly) one third-M.SG of-DET.F.PL wall-F.PL COP.3SG covered.M.SG by mold
 $\checkmark M, \times C$
- b. (esattamente) un **terz-o** del-le **paret-i** sono coperte di muffa
 (exactly) one third-M.SG of-DET.F.PL wall-F.PL COP.3PL covered.F.PL by mold
 $\times M, \checkmark C$

- I should mention at this point that the measure interpretation is compatible with other scenarios as well; e.g., one where the lower third of every wall is covered in mold
- I use the above dichotomy for mostly expository purposes
- This pattern extends to other kinds of nouns and other dimensions of measurement

- (4) ‘A third of the films were watched by John’ [Italian]

- a. **un terzo** de-i film è stato visto
 one third of-M.PL film COP.3SG been.M.SG watched.M.SG
 $\checkmark M, \times C$
- b. un **terzo** de-i **film** sono stati visti
 one third of-M.PL film COP.3SG M.PL watched.M.PL
 $\times M, \checkmark C$

- This pattern generalizes to a certain extent to a variety of other Indo-European languages
- See French, Greek, which replicate our findings for Italian

- (5) ‘One third of the walls are covered by mold’ [French]

- a. **un tiers** des murs est couvert de moisissures
 one third of.DET.M.PL wall.M.PL COP.3SG covered.M.SG by mold
 $\checkmark M, \times C$
- b. un **tiers** **des** **murs** sont couverts de moisissures
 one third of.DET.M.PL wall.M.PL COP.3PL covered.M.PL by mold
 $\times M, \checkmark C$

- (6) ‘One third of the pies were eaten’ [Greek]

³Reminder: In $\frac{x}{y}$, x is the numerator and y is the denominator.

- a. **ena trit-o** ton pit-on fagoθik-e
 one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3SG.PST
 ✓M, ×C
- b. **ena trit-o ton pit-on** fagoθik-an
 one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3PL.PST
 ×M, ✓C

- A similar pattern can be observed with the fraction partitives of Russian and German, albeit with the caveat that a count reading is still available when Agreement is with the denominator

(7) ‘One third of these⁴ cakes were eaten’ [Russian]

- a. **(odna) tret’** étix pirog-ov byl-a s’’eden-a
 one third.F.SG this.GEN.PL cake-GEN.PL be-F.SG eaten-F.SG
 ✓M, ✓C
- b. **(odna) tret’ étix pirog-ov** byl-i s’’eden-i
 one third.F.SG this.GEN.PL cake-GEN.PL be-PL eaten-PL
 ×M, ✓C

(8) ‘One third of the cakes were eaten’ [German]

- a. **Ein Drittel** der Kuchen wurde aufgegessen
 one third DET.GEN.PL cake.GEN.PL be.PST.3SG eaten
 ✓M, ✓C
- b. Ein Drittel **der Kuchen** wurden aufgegessen
 one third DET.GEN.PL cake.GEN.PL be.PST.3PL eaten
 ×M, ✓C

- Again compare American English—a language which lacks the measure reading altogether

(9) One third of the walls {*is/are} moldy
 ×M, ✓C

- Much prior work on FPs posits a DP-within-DP structure; under a theory of Agree where locality/headedness resolves goal competition (Chomsky, 1995), agreement with the FP complement at all is independently hard to explain
- **How do we make sense of this and the semantic shifts which come with it?**
- **Objective:** provide an account of the phenomena while also explaining the typological facts

2 Prelude: Syntactic vs. Semantic Agree

- For this problem, there are two (broad) notions of agreement and how they bear on interpretation
- **I. Syntactic Agreement:** the realization of a feature in multiple locations given some structural configuration (i.e., the Agree we all know and love)
- As I need not explain to this audience, this notion is formalized a few different ways in syntactic theory
 - The PROBE-GOAL machinery of the Minimalist Program and the many permutations which following works propose
 - Agreement via SPECIFIER-HEAD relations

⁴Consultant found the measure reading easier to attain with a demonstrative heading the complement. The measure reading is still marginally available without a demonstrative.

- To derive the two agreement options for FPs, we must either assume two distinct structures which generate different results for Agree, or tweak the rules of feature-copying/Spell-Out to get two bundles of features from a single structure

II. **Semantic Agreement:** features on external heads (T/Infl) are not uninterpretable copies but are actually presuppositions relevant to a nominal ([Sportiche and Charnavel, 2025](#))

- Devised to account for agreement optionality like that in (10), which also comes with an interpretational alternation

(10) The committee {has/have} come to a decision [British English]

- The superficial similarity of these classic semantic agreement cases and the topical alternation might motivate us to pursue an analysis similar to Charnavel & Sportiche
- It is especially attractive considering it would permit us to avoid positing structural differences between measure and count FPs—revisit the data in §1 and you will see that FPs look the same regardless of what readings are available
- Unfortunately, this mode of thinking doesn't get us far for the current puzzle
- Crucially, both readings are accessible if there is no agreement with the fraction—see (2), also (11)

(11) o Giánnis éfage ená trít-o ton biskóton
the Yiannis ate.3SG one third-N.SG the cookie.GEN.N.PL
'Yiannis ate one third of the cookies' [Greek]
✓M, ✓C

- AFAICT, it is impossible to get additional readings outside of positions of agreement since the features on the agreeing head licenses the interpretational shift
- Furthermore, a C&S-type account is successful only on a singular-plural agreement contrast
- This is a mischaracterization of the topical data, though: the meaning is conditioned by whether the agreeing features match those on the denominator or the complement
- We know this isn't about singular-plural agreement because in many of these languages, singular agreement becomes impossible if the numerator > 2

(12) *weil zwei Drittel der Student-en unzufrieden ist
because two third DET.GEN.PL student-GEN.PL discontent be.3SG
Int. 'Because two thirds of the students are unhappy...' [German]

(13) ??duo trít-a ton maθíton adipaθi ton Yani
two third-N.SG the student.GEN.PL dislike.3SG the Yiannis
'Two thirds of the students dislike Yiannis' [Greek]

(14) deux cinquièmes des étudiants {*/a/ont} lu War & Peace
two fifth.M.PL of.DET.M.PL student.M.PL have.{3SG/3PL} read War & Peace
'Two-fifths of the students have read War & Peace' [French]

- Given these facts, I turn away from a C&S-inspired analysis and instead opt for an analysis where the agreement facts are due to syntactic processes
- Specifically, I assume the target of agreement is determined by a distinct structure—by the tenet of ONE FORM/ONE MEANING, I further posit that both structures have distinct LFs
- Independent of semantics, structural ambiguity for FPs is something we may want anyway—this will be revisited in §4
- How exactly each structure gives us the appropriate reading constitutes the bulk of this talk

3 Anatomy of a Fraction Partitive

- I take this section to more-closely examine the construction at hand and identify some properties which may be relevant to an analysis
- Generally, a fraction partitive in the sampled languages can be broken into two parts: the fraction itself and the complement
- Mathematically, fractions are represented with two numbers: a NUMERATOR and a DENOMINATOR, the value of the fraction being the former divided by the latter
- The languages considered in this study (typically) denote a fractional number through a combination of numerator and denominator⁵
- While the numerator slot is filled by a cardinal number, the denominator takes a special form of the numeral
- In the languages considered, the form of the denominator is often derived from or homophonous with the number's corresponding ordinal—not a universal, but a very strong cross-linguistic generalization (David Pesetsky, p.c.)

Language	Ordinal	Denominator
English	fifth	fifth
Italian	quint-	quinto
French	cinquième	cinquième
Greek	péempt-	péempto
German	fünfte	fünftel
Russian	pjat-	pjataja

Table 1: The number 5 in its ordinal and denominator forms by language

- This of course raises the question of whether we should derive the semantics of fractions from those of ordinals; [López Palma \(2011, 2012\)](#) is one author who analyzes them as such
- Since it does not bear on the puzzle we are exploring here, I will not go into detail about it at this juncture
- In many of these languages, what appear to be number and (if applicable) gender features are visible on the denominator
- Specifically, the plural feature is licensed if the numerator > 1

- (15) a. one fifth
 b. two fifths

⁵This is not the case for all natural languages, though ([Anicotte, 2022](#)); Coptic is one such case. I thank Emma Glenister for bringing this to my attention.

- (16) [Italian]
 a. un terz-o
 a third-M.SG
 b. due terz-i
 two third-M.PL

- (18) [Greek]
 a. éna trít-o
 one third-N.SG
 b. dúo trít-a
 two third-N.PL

- (17) [French]
 a. un cinquième
 a fifth.M.SG
 b. deux cinquième-s
 two fifth.M-PL

- (19) [Russian]⁶
 a. (odna) vos'm-aya
 one eighth-NOM.F.SG
 b. sem' vos'm-ikh
 seven eighth-GEN.PL

- German is an interesting exception to this generalization, but the agreement facts in (12) tell us that it is not agreed with like a singular noun

- (20) {ein/sieben} Zehntel
 {one/seven} tenth

- In some of these languages with a gender distinction, the gender of denominators is not homogenous
- French *moitié* and German *Hälfte* 'half' are exceptionally feminine despite other denominators exhibiting masculine and neuter case, respectively (Lederer, 1977)

Language	Denominator γ
Italian	MASC
French	MASC, FEM
Greek	NEUT
German	NEUT, FEM
Russian	FEM

Table 2: Gender(s) of denominators by language

- Note that not all partitive heads have inherent features; in numeral partitives in Greek, for example, a complex numeral containing 'one' agrees with the complement

- (21) 'Thirty-one of the {men/women}' [Greek]
 a. triánta énas após tous ándres
 thirty one.M.PL of the.M.PL men
 b. triánta mía após tis gunaíkes
 thirty one.F.PL of the.F.PL women

- If the appropriate licensing conditions are met, denominators can also take definite articles, demonstratives

- (22) to éna trít-o
 DET.M.SG one third-M.SG
 'The one third' [Greek]

- (23) a mozhët eta tret' byla ne toj
 CONJ maybe this.F.SG third was.F.SG NEG that.F.SG
 'Or maybe this third wasn't the right one?' [Russian]⁷

⁶Fractions with a numerator greater than one receive genitive case (Timberlake, 2004, p. 194). As far as I know, this is separate from the paucal genitive phenomenon observed in Russian and other Slavic languages (see Pesetsky 2013 and works cited therein).

⁷From Grigory Leps - *Pannjaja tišina* (Early Silence). <https://genius.com/Grigory-leps-early-silence-lyrics>

- If there is no overt numerator, it's value is inferred to be 1
- Furthermore, fractions can be modified by adjectives and relative clauses

(24) ein ungebildetes Drittel von uns
 one uneducated.NOM.SG third from 1PL.ACC
 'An uneducated third of us' [German]

(25) tret' studént-ov kotoraja včera xodil-{a/*i} v-bar
 third student-GEN.PL which.SG yesterday went-{SG/PL} in-bar
 'The third of the students who went to the bar yesterday' [Russian]

(26) Il terz-o del-le paret-i dipint-o da John
 DET.M.SG third-M.SG of-the wall-F.PL painted-M.SG by John
 'The third of the walls painted by John' [Italian]

(27) le tiers des étudiants auquel Martin a fait cours
 DET.M.SG third of.DET.M.PL student-PL who.M.SG Martin have.3SG made class
 'The third of the students that Martin has taught' [French]

- This is all to say that in the languages surveyed, fractions look pretty nominal-ish
- As for the complement, it is introduced with a preposition or marked with genitive case

(28) a. due terz-i del-le arance
 two third-M.PL of-the orange
 'Two thirds of the orange' [Italian]

b. un sixième de la pomme
 one sixth of the apple
 'One sixth of the apple' [French]

(29) syem' voc'myerok jabloka
 seven eighths apple.GEN
 'Seven eighths of the apple' [Russian]

- In at least German and Greek, both options are available; as far as I am aware, the choice of the preposition or genitive case has no bearing on the alternation I explore in this study

(30) zwei Drittel {des Apfels / vom Apfel}
 two third {the.GEN apple.GEN / of.the.DAT apple.DAT}
 'Two thirds of the apple' [German]

(31) to éna ékto {apó to mίlo / ton mίlon}
 the one sixth {of the.ACC apple.ACC / the.GEN apple.GEN}
 'One sixth of the apple' [Greek]

- Up to this point, I've been talking about FPs which contain a plural DP; however, singular complements are also possible
- In some of these languages, there is still agreement optionality for FPs with a singular complement

(32) deux tiers de la surface sont aménagés dans un esprit lounge
 two third.PL of the surface be.3PL fitted.out in a style lounge
 'Two thirds of the surface is outfitted in a lounge style' [French]

(33) deux tiers de la biodiversité mondiale vit dans le sol
 two third.PL of the biodiversity on.earth live.3SG in the soil
 'Two thirds of Earth's biodiversity lives in the soil' [French]

(34) dio trit-a tu mil-u ine kokin- $\{o/a\}$
 two third-NEUT.PL DET.GEN.SG apple-GEN.SG COP.3 red- $\{NOM.N.SG/NOM.N.PL\}$
 ‘Two thirds of the apple is red’ [Greek]

(35) weil zwei Drittel einer Birne essbar $\{sind / ist\}$
 because two third a.GEN.F.SG pear.GEN.SG edible $\{be.3PL / 3SG\}$
 ‘Because two thirds of a pear is edible’⁸ [German]

- Some languages, though, can only have Agreement with the denominator if the complement is singular

(36) due terz-i del-la mela $\{ \checkmark \text{ sono ross-i} / *è \text{ rossa} \}$
 two third-M.PL of-DET.F.SG apple $\{ be.3PL \text{ red-M.PL} / be.3SG \text{ red-F.SG} \}$
 ‘Two-thirds of the apple are red’ [Italian]

(37) dve tretí étego piroga s’edaj-em- $\{y/*\emptyset\}$ za tri minuty
 two thirds this.GEN.SG cake.GEN.SG eat-PTCP- $\{PL/SG\}$ in three minutes
 ‘One can eat two thirds of this cake in three minutes’⁹ [Russian]

- American English displays the reverse behavior—Agreement can be either with the fraction or denominator if the complement is singular.

(38) Two-thirds of the eclair was/ $\%$ were already eaten by the time George found it

- Table 3 shows all of the Agreement possibilities we’ve discussed so far

Langs.	SG Comp.		PL Comp.	
	Frac.	Comp.	Frac.	Comp.
Am. English	$\% \checkmark$	$\checkmark$	$\times$	$\checkmark$
German	$\% \checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Italian	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
French	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Greek	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Russian	$\checkmark$	$\times$	$\checkmark$	$\checkmark$

Table 3: Agreement possibilities by number of complement, by language.

- I don’t strive to develop a theory for the agreement under singular complements in this work; however, I aim to have my analysis not preclude an explanation of the singular complement facts

4 Syntactic Correlates of Agreement

- Putting aside the interaction of agreement and semantics, agreement has effects for syntax as well
- For example, the presence of nominal modifiers (RRCs, adjectives, etc.) on a fraction “block” agreement with the complement

(39) ein ungebildetes Drittel von uns $\{hat/*haben\}$ für ihn gestimmt
 one uneducated.NOM.SG third from 1PL.ACC have- $\{3SG/1PL\}$ for 3SG.M.ACC voted
 ‘An uneducated third of us voted for him.’ [German]

(40) [tret’ studént-ov kotoraja včera xodil- $\{a/*i\}$ v-bar] sobral- $\{as’/*is’\}$ snova
 [third student-GEN.PL which.SG yesterday went- $\{SG/PL\}$ in-bar] gather- $\{SG/PL\}$ again
 ‘The third of the students who went to the bar yesterday gathered again’ [Russian]

⁸Note that some speakers of German cannot have Agreement with denominator if the complement is definite.

⁹Default Agreement (N.SG) is OK in some circumstances.

(41) krasnaja tret' stulje-v stojal- $\{a/*i\}$ v-uglu
 red.NOM.F.SG third chair-GEN.PL stand- $\{SG/PL\}$ in-corner
 'The red third of the chairs are standing in the corner' [Russian]

(42) Il [terz-o del-le paret-i dipint-o da John] $\{\grave{e}$ ross-o / *sono
 DET.M.SG [third-M.SG of-the wall-F.PL painted-M.SG by John] $\{be.3SG\}$ red.M.SG / be.3PL
 ross-e}
 red-M.PL}
 'The third of the walls painted by John are red' [Italian]

(43) 'The third of the students that Martin has taught have studied semantics' [French]
 a. le tiers des étudiants auquel Martin a fait cours a
 DET.M.SG third of.DET.M.PL student-PL who.M.SG Martin have.3SG made class have.3SG
 étudié la sémantique
 studied DET.F.SG semantics
 b. *le tiers des étudiants auxquels Martin a fait cours ont
 DET.M.SG third of.DET.M.PL student-PL who.M.PL Martin have.3SG made class have.3PL
 étudié la sémantique
 studied DET.F.SG semantics

- Bizarrely, though, American English does not have this problem

(44) The [one third of the students who like John] also **like** Mary.

- Some of these languages have a similar effect with definite articles

(45) èta tret' studént-ov sobral- $\{as'/*is'\}$
 this.F.NOM.SG third student-GEN.PL gather- $\{SG/PL\}$
 'This third of the students gathered' [Russian]

(46) to ena trito ton milon ine $\{kokino/?kokina\}$
 DET.N.SG one third DET.GEN.PL apple.GEN.PL COP red. $\{N.SG/PL\}$
 'One third of the apples are red' [Greek]

- See also evidence from Binding; the features of a reflexive pronoun is determined by the part of the anaphor that is agreed with

(47) ena trit-o ton mathit-on_i ið-an ton eafto_i tus
 one third-N.SG DET student-GEN.M.PL see-3PL.PST DET self.ACC GEN.M.PL
 'One third of the students saw themselves' [Greek]

(48) *ena trit-o ton mathit-on_i ið-e ton eafto_i tus
 one third-N.SG DET student-GEN.M.PL see-3SG.PST DET self.ACC GEN.M.PL
 Intended: 'One third of the students saw themselves' [Greek]

(49) ena trit-o ton mathit-on_i ið-e ton eafto_i tu
 one third-N.SG DET student-GEN.M.PL see-3SG.PST DET self.ACC GEN.N.SG
 'One third of the students saw themselves' [Greek]

- PP-extraposition precludes agreement with the complement in at least Greek

(50) ena trít-o ___ ine kokin- $\{o/*a\}$ apo ta mila
 one third-M.SG COP.3 red- $\{M.SG/PL\}$ of the apple.PL
 'One third of the apples are red' [Greek]

- The evidence put forth in this section bolsters the argument I made in §2; despite superficial similarities, measure and count FPs actually correspond to separate structures

5 Deriving the Count Reading

- I take this section to consider the “canonical” way that fraction partitives are analyzed (i.e., how to get the count interpretation)
- Drawing heavily from [Wagiel \(2021\)](#) here, but other authors present related ideas
- Descriptively, there are two components to an FP’s semantics...
 - parthood: the referent of an FP must be a proper part of the complement inside the FP
 - comparison of degrees: the measure of the referent on a scale is a fraction the measure of the complement on the same scale
- Recall that FPs can take both singular and plural individuals; this generalization can be observed across many partitives in many languages (observed also by [Moltmann 1997](#))
- This is true even of languages like Passamaquoddy, which is very much unlike the Indo-European languages we’ve been considering so far

- (51) a. epahsiw cikon wekine
 half apple.PROX.SG spoil.AI
 ‘half of the apple spoiled’
 b. epahsiw cikoni-yik wekini-yik
 half apple-PROX.PL spoil.AI-PROX.PL
 ‘half of the apples spoiled’ [Passamaquoddy]

- Under [Link \(1983\)](#), who posits two relations for parthood, this is hard to make sense of
 - ATOMIC part-of: holds between pluralities and the (sums of) individuals which compose them
 - MATERIAL part-of: holds between singular individuals and their parts
- Given this, Moltmann and Wagiel collapse Link’s two parthood relations into one ($\sqsubset$)
- Doing so makes a seemingly problematic prediction, though; recall that parthood is defined as a transitive relation ([Leonard and Goodman, 1940](#))

- (52) a. $x \sqsubset y \wedge y \sqsubset z$
 b. $x \sqsubset z$

- If Jerry’s hand is part of Jerry and Jerry is part of the New Yorkers, it should follow that Jerry’s hand is part of the New Yorkers
- Yet Jerry’s hand being covered in paint is clearly not a verifying condition for (53)

(53) Part of the New Yorkers are covered in paint

- Proposed solution: Transitivity of parthood is obviated if there is an intervening individual which has the property of being an INTEGRATED WHOLE—Moltmann’s TRANSITIVITY PRINCIPLE, Wagiel’s TOPOLOGY-SENSITIVE TRANSITIVITY

(54) $\forall x, z. x \sqsubset z \rightarrow \neg \exists y. x \sqsubset y \wedge y \sqsubset z \wedge \text{int-wh}(y)$

- How they define integrated whole varies (Moltmann appeals specifically to R-integrated wholes ([Simons, 1987](#)) while Wagiel captures the property with MAXIMALLY-STRONG SELF-CONNECTEDNESS ([Casati and Varzi, 1999](#)))
- The end result is the same, though—a materially-real, single, whole individual like *Jerry* constitutes an integrated whole whose parts are not members of any plurality the integrated whole is part of
- I call this analysis of parthood FORMALIZED TRANSITIVITY SUSPENSION (FTS)
- So now we have a parthood component which can handle singular and plurals DPs; how about comparison?
- Wagiel assumes variable measure functions (represented μ) which are defined by context and the properties of the MF’s argument
- This flexibility is desirable because (a) FPs can take many kinds of individuals, not all of which can be measured in the same way, and (b) even the same individual can be measured distinct ways in an FP

- (55) a. Paint half of the apple $\rightsquigarrow$ surface area
 b. Use half of the apple for the pie $\rightsquigarrow$ volume, mass

- However, Wagiel argues that plurals are always counted rather than measured
- He defines fractions as in (56)

$$(56) \quad [[\text{third}]] = \lambda y_e \lambda x_e . x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3}$$

- The identity of any given μ is chosen from a set of all non-monotonic measure functions I’ll call M
- M has a partial ordering s.t. cardinality is minimal with respect to every other measure function

$$(57) \quad M = \{\mu_{card} < \mu_{vol}, \mu_{runtime}, \dots\}$$

- Given this configuration for measure functions, Wagiel adapts an algorithm for defining measure functions from [Bale and Barner \(2009\)](#)

(58) Contextual conditioning
 μ is interpreted as one of the measure functions m_z in the series $\langle m_1, m_2, m_3, \dots, m_n \rangle$ such that the argument for μ is in the range of m_z ; furthermore, contextually m_z is preferred to m_y if $m_z < m_y$

- Only plural individuals can be the input of a cardinality function, so cardinality is guaranteed to arise if μ takes a plural individual (see [Wellwood 2019](#) for a similar idea)¹⁰
- The count reading follows because of the ordering of the set in (57)

6 Previous Attempts at Deriving Measure

- The existence of a measure reading for fraction partitives is not a wholly original observation; something like it was first mentioned in [Hoeksema \(1996\)](#), AFAIK, and followed up by [de Hoop \(1997, 2003\)](#); [Falco and Zamparelli \(2019\)](#)¹¹

¹⁰Wagiel does not entertain whether a variable MF can be anything else given a plurality. Bale & Barner, who specifically focus on object mass nouns (e.g., *furniture*) set up variable MFs so that noncardinal are not necessarily excluded given a plurality (see also [Barner and Snedeker 2005](#)). Their system is not deterministic, but Wagiel (AFAICT) never gives a reason why his shouldn’t be. I think Wellwood’s algorithm for defining MFs is not at the whim of context in this way.

¹¹You may notice in these texts that these explorations of the interpretation of FPs cite English data. I personally disagree with the judgments for my own English and maintain that the measure reading for FPs is not robustly attested in American English. As far as I am able to tell, none of Hoeksema, de Hoop, Falco, or Zamparelli are native American English speakers, and I have shown that the measure/count alternation is quite common in the Indo-European languages spoken throughout continental Europe. These factors make me suspect that judgements from their respective L1s are influencing how they can interpret these FPs.

- The fact that the reading is licensed by a particular agreement configuration went unnoticed, though
- As a first attempt at deriving measure FPs, let us consider a suggestion from Falco & Zamparelli and extend the syntax to get the appropriate agreement
- de Hoop also provides an account of the measure interpretation, essentially resolving the two meanings to the semantic type of the complement (e or $\langle e, t \rangle$)
- I do not go into the details here for the sake of time
- Why is the measure reading ungenerable under the account in §2?
 - Transitivity suspension: parts of individuals in a plurality are not themselves part of the plurality
 - Bias to counting: Because μ defaults to counting if it's argument is a plurality, there is no other way to measure it
- If we want to hold these constant, one approach would be to change something about the plural structure of the complement—i.e., “trick” μ into thinking that it's argument is singular/uncountable
- F&Z's suggestion: a covert operator takes a plural individual and returns unindividuated mass—a UNIVERSAL GRINDER (Pelletier, 1975)
- After all, a mass noun cannot be counted and has no atomic/IW-ish parts
- My attempt below: a grinder which takes an individual and returns the generalized sum of the parts of it's proper parts

$$(59) \quad [[UG]] = \lambda x. \sqcup \{z : \forall y. z \sqsubseteq y \wedge y \sqsubset x\}$$

- AFAIK, the UG was originally designed to take a predicate rather than an individual, so I had to get creative here
- Without plural structure, there is nothing for μ to count—interpreted as one big object
- Importantly, wall-parts and their sums are now part of $UG([[le\ pareti]])$ (Thanks, transitivity suspension!)
- Explaining Agreement: In a relativized Agreement system like Deal (2015), a probe will prefer to Agree with something not mass-y, instead copying the features on the fraction
- Prediction: If a fraction partitive contains a regular (read: not derived by grinder) mass noun, anything agreeing should agree with the fraction only
- Problem: the above is not substantiated

(60) zwei Drittel des Schnees {ist / sind} eisig
 two thirds DET.GEN.SG.M snow.GEN {COP.3SG / COP.PL} icy
 ‘Two thirds of the snow is icy’ [German]

(61) zwei Drittel des Wassers {ist / sind} vergiftet
 two thirds DET.GEN.SG.M water.GEN {COP.3SG / COP.PL} poisoned
 ‘Two thirds of the water is poisoned’ [German]

(62) duo trita ton çumos ine pradin-{a/os}
 two third.N.PL DET.GEN.M.SG juice.GEN COP green-{N.PL/M.SG}
 ‘Two thirds of the juice is green’ [Greek]

- We'd have to derive agreement some other way; perhaps it is UG blocking agreement rather than some feature of mass nouns?

- Also, observe that in English coercion of a count noun to a mass noun comes with a change in morphology; massified count nouns are morphologically singular and do not need an article

(63) **Three squirrels** carelessly wandered into traffic and were struck by a motorist.
Now there is **squirrel** all over the road.

- If F&Z are right, should we expect something similar for the complement FP under the measure reading?
- More damningly: distributivity and distributive interpretations are still possible with measure FPs

(64) to ena trito ton vivlion zugiz-ei 4kg to kaθena
DET.NOM.SG one third DET.GEN.PL book.GEN.PL weigh-SG 4kg DET each.N
'One third of the books weigh 4kg each' [Greek]

(65) 'A fifth of the boys moved the piano' [French]
a. un cinquième des garçons a bougé un piano
a fifth of.DET.PL boy.PL have.3SG moved a piano
✓ DISTRIBUTIVE, ✓ COLLECTIVE
b. un cinquième des garçons ont bougé un piano
a fifth of.DET.PL boy.PL have.3PL moved a piano
✓ DISTRIBUTIVE, ?? COLLECTIVE

- See also individual-level predicates

(66) ena trit-o ton maθit-on ine eksipn-{o/i}
one third-NEUT.SG DET.GEN.PL student-GEN.PL COP.3 smart-{NOM.NEUT.SG/NOM.MASC.PL}
'One third of the students are smart' [Greek]

(67) ena trit-o ton maθit-on ine arost-{o/i}
one third-NEUT.SG DET.GEN.PL student-GEN.PL COP.3 sick-NOM.NEUT.SG/-NOM.MASC.PL
'One-third of the students are sick' [Greek]

(68) Un terzo del-le ragazz-e {é andat-o / sono andat-i} in spiaggia
One third of-DET.F.PL girl-F.PL {have.3SG walked-M.SG / have.3PL walked-F.PL} to beach
'One third of the girls walked to the beach' [Italian]

- Compare unambiguously massified DPs

(69) An army of Boston Dynamics robots were smashed to pieces by the anti-AI protesters.
#Now robot is (each) begging for mercy.

- If this solution were right, we would expected downstream effects—but there don't seem to be any
- The key takeaway from this section is that measure FPs can (but do not necessarily) denote pluralities, making this kind of approach insufficient

7 Russian Interlude

- Falco & Zamparelli (and also de Hoop) place the burden of the interpretational change on the complement of the FP; i.e., there is something invisible which gives us the measure reading over the count reading
- This does not get us very far for the reasons I list in §6; furthermore, how do we explain the differences in syntactic behavior discussed in §4?
- **New direction:** what if the element which gives us the measure reading is actually hiding in the fraction field rather than the complement

- Evidence from Russian supports for this avenue; on top of the “bare” FPs we’ve seen up to this point, there are also FPs which include the part-word *čast’*
- In FPs where *čast’* is present, **agreement with the complement is blocked and only the measure reading is available**

(70) tret’ja **čast’** sten krasn- $\{\checkmark \text{ aja/*ye}\}$
 third.F.SG part.NOM.F.SG wall.GEN.F.PL red- $\{\text{F.SG/PL}\}$
 ‘A third part of the walls are red’
 $\checkmark \text{M}, \times \text{C}$

- I take this item to be the overt realization of a normally-covert operator which shifts the denotation from count to measure
- This idea comes with a big qualifier: I cannot take this use of *čast’* to be ‘part’, despite appearances
- The referents of measure and count FPs are both parts of their respective complements; *čast’* does not add any non-redundant *part*-ishness to the FP
- Additionally, some languages have constructions superficially similar to Russian FP-*čast’* yet do not exhibit analogous effects; English is such a language, in fact

(71) [A] third part of the waters **were** turned into wormwood (COCA, [Davies 2010](#))

- Regardless, the effect of *čast’* on agreement and interpretation is not negligible; thus, I pursue the possibility that it means something other than ‘part’ in this construction

8 Proposal

8.1 TOTAL, the (occasionally overt) operator

- I aim to reduce the measure/count contrast to the presence or absence of a special operator TOTAL
- By default, the count reading will emerge (per [Wagiel 2021](#), kind of) and the complement will be agreed with
- Introducing TOTAL manipulates the syntax and semantics to get the measure reading and its corresponding agreement
- I reason that TOTAL can be optionally overt in some languages, like Russian (where it is realized as *čast’*), but it is usually covert
- This way, difference between American English and other languages is a lexical one: Greek, French, etc. have it, English does not

8.2 Semantically

- To get this analysis off the ground, we need to make two changes to Wagiel’s proposal: no more transitivity suspension, and shiftability of a measurement parameter

8.2.1 Part is just part

- I maintain that $\sqsubset$ captures cases of both material part and atomic part (in the sense of [Link 1983](#))
- However, I deviate from Wagiel (and also [Moltmann 1997](#)) in insisting upon parts of integrated wholes being parts of pluralities
- While once the set denoted by (72) included only “atomic” walls and their sums, now wall-parts and their sums are included

(72) $\lambda x.x \sqsubset [[\text{le pareti}]]$

- ...but we do not want to throw away the badness of things like (73)

(73) ??/#Cooper's hand is (a) part of the linguists

- Suggestion: perhaps transitivity suspension is actually a pragmatic effect, the result of competition
- This can be illustrated with the maxim in (74)

(74) Parthood-Subset Maxim
 $a \sqsubset b \rightarrow \{x : x \sqsubset a\} \subset \{y : y \sqsubset b\}$

- Assuming a gunky universe (one where every individual has a proper part), both sets will have a cardinality of ∞
- Yet, a 's parts will constitute a proper subset of b 's parts because b contains a (and thus a 's parts are included in the set of b 's parts)
- Its relevance to the issue is particularly clear if we imagine a as a singular individual like *Cooper* and b as a plurality of which a is a member (*the linguists*)
- On its literal interpretation *part of Cooper* is far more informative than *part of the linguists*
- By conversational reasoning à la [Grice \(1975\)](#), if you want to refer to part of an atomic individual, you should use something singular
- Therefore, I reason that the judgment that *part of the linguists* cannot refer to my hand is an implicature and not something encoded by the grammar
- Prediction: in cases of epistemic uncertainty, implicature should be obviated (cf. plurals, [Sauerland et al. 2005](#))
- At least in English, this is realized: see questions, universal quantifiers

(75) a. Has John painted part of the boats?
 b. Yes, in fact he painted the starboard side of the USS Bismarck.

(76) Every boy painted part of the boats. John painted part of the USS Bismarck, Paul painted the USS Maine and USS Eisenhower, ...

- See also (77), which seems to be OK if we do not know which boat the winch came from

(77) This winch {must be/is} part of Columbus' boats (but I'm not sure which one)

- We also observed homogeneity-like effects; given the definition in (78), the contradiction in (79) should not appear if a sub-ship part is not in the set denoted by the partitive
- This fact and the acceptability of the cases above cannot be explained by Moltmann or Wagniel's proposals

(78) $[[\text{no}]] = \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle} . \neg \exists x . P(x) \wedge Q(x)$

(79) No part of the boats {was/were} painted. #Jerry (only) painted half of USS Ticonderoga.

- This data shows that plural partitives *can* mean part of an individual in that plurality; this is transparent when certain felicity conditions are met
- I'll expand upon this idea in my Elsewhere talk later this month, so stop by if you want to hear more!

- **What does this mean for the current puzzle?** Now, fraction partitives can refer to parts of the plurality which are not integrated wholes or sums of integrated wholes (without deferring to something like a universal grinder)

8.2.2 M parameter

- Recall the partially-ordered set of measure functions M by which variable measure functions are defined

$$(80) \quad M = \{\mu_{card} < \mu_{vol}, \mu_{mass}, \dots\}$$

- I treat M as a parameter which, like other parameters, can be shifted via intensional operators—TOTAL is the guilty party for this case
- Given the ordering for MFs and [Bale and Barner \(2009\)](#); [Wellwood \(2019\)](#); [Wagiel \(2021\)](#)'s rules for defining measure functions, count will always emerge if a given individual is countable (i.e., plural)

$$(81) \quad [[\text{terzo}]]^M = \lambda y_e \lambda x_e.x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3}$$

$$(82) \quad [[\text{terzo delle pareti}]]^M = \lambda x_e.x \sqsubset \max(\lambda y_e.*\text{wall}(y)) \wedge |x| = |\max(\lambda y_e.*\text{wall}(y))| \times \frac{1}{3}$$

- *Pump the brakes!!* Partitives can now denote any part of a plurality; we also know that rational numbers appear in natural language, see (83)

$$(83) \quad \text{John read } \{2.5/\text{two and a half}\} \text{ books}$$

- Could the measure function return a non-integer cardinality? This doesn't get us the count reading!
- Counterpoint: nonintegers like 2.5 are complex, the sum of a counting function as well as a partiality function ([Haida and Trinh, 2021](#))
- Given the syntactic complexity of these kinds of numeral constructions, it may also lend itself to an analysis like that in [Ionin and Matushansky \(2006\)](#)
- It seems very reasonable to conclude that cardinality functions will only return integers
- **Measure reading:** TOTAL shifts the “default” M parameter to M' , which corresponds to a different ordering for the set of MFs (cardinality is outranked by other forms of measurement)

$$(84) \quad [[\text{TOTAL}]] = \lambda P. [[P]]^{M'} \text{ where } M' = \{\mu_{vol}, \dots < \mu_{card}\}$$

- (For German, we might say that in this language TOTAL just removes the ordering rather than adjusts it, since count and measure interpretations are both available with frac. agreement)
- And so the measure reading follows

$$(85) \quad [[\text{TOTAL terzo delle pareti}]]^{M'} = \lambda x_e.x \sqsubset \max(\lambda y_e.*\text{wall}(y)) \wedge \mu(x) = \frac{\mu(\max(\lambda y_e.*\text{wall}(y)))}{3}$$

8.3 Syntactically

- Syntactically, there's two possible ways the agreement facts can play out:

- I. Only TOTAL bears ϕ -features; the fraction words we see do not genuinely have features (see [Klockmann 2018](#) on pseudopartitives)

$$(86) \quad [\text{a third}_{\phi:\emptyset} [\text{the walls}_{\phi:\text{PL}}]] \quad \text{COUNT}$$

$$(87) \quad [\text{a third}_{\phi:\emptyset} [\langle \text{TOTAL} \rangle_{\phi:\text{SG}} [\text{the walls}_{\phi:\text{PL}}]]] \quad \text{MEASURE}$$

- So what explains the apparent number/gender morphology we see on denominators?
- Perhaps this morphology is a kind of concord with this TOTAL
- Immediate problem: number suffixes are always present on the fraction, even when presumably there would be no TOTAL
- Furthermore, we saw in some languages (French and German) that denominators can vary in gender; agreement follows suit for the exceptional denominators

(88) ‘Half the cake is moist’ [French]

- | | | | | | | |
|----|----------|-----------|-------------|-----------|---------|------------|
| a. | la | moitié | des | gâteaux | est | moelleuse |
| | DET.F.SG | half.F.SG | of.DET.M.PL | cake.M.PL | COP.3SG | moist.F.SG |
| b. | la | moitié | des | gâteaux | sont | moelleux |
| | DET.F.SG | half.F.SG | of.DET.M.PL | cake.M.PL | COP.3PL | moist.M |
| c. | *la | moitié | des | gâteaux | est | moelleux |
| | DET.F.SG | half.F.SG | of.DET.M.PL | cake.M.PL | COP.3SG | moist.M |

- Possible out: In French, there are two TOTAL operators with different genders—this feels stipulative

II. Agreement corresponding to the measure reading is genuinely agreement with the denominator—syntactic circumstances affect whether these features are accessible

- Count reading (no TOTAL): complements are the real heads of FPs (inspired by [Selkirk 1977](#))
- In languages where fraction and complement are delimited by a preposition, we will have to concede that this item is not projecting any structure—maybe it is a case-marker ([Chomsky, 1970](#)) or a late-merged element ([Chomsky, 1981](#))
- By feature percolation, the features of the complement are more local to probes (and thus complement agreement follows)

(89) [DP:F.PL [NP? un terzo_{M.SG}] [DP (del) le pareti_{F.PL}]]

- I analyze *un* and translational equivalents not as an indefinite article but as numeral 1; this keeps it in the NP field rather than DP, which we would need for agreement facts
- This interpretation is plausible given the cross-linguistic tendency for indefinite articles to be diachronically derived from ‘one’ ([Givón, 1981](#))
- To get the measure reading, TOTAL takes the DP in (89)
- In addition to its update to the denotation, TOTAL comes with an uninterpretable feature that it needs to check—let’s call it η
- Additionally, fractions bear an interpretable η and can thus satisfy TOTAL
- The operator will look through its c-command domain to find a phrase with this feature and move it
- Via projecting movement, the label of this NP projects ([Bhatt, 2002](#))
- This yields a configuration where the fraction NP’s ϕ -features are structurally higher than those of the internal DP
- Now Agreement with the fraction is possible

(90) [DP:M.SG [NP un terzo_{M.SG} [XP TOTAL [DP:F.PL [NP ~~un terzo~~_{M.SG}] [DP le pareti_{F.PL}]]]]]]

- Secondary result: this is the structure we would need for relative clauses, definite articles, and PP displacement (see §4) we observe with fraction agreement in some languages

References

- Anicotte, R. (2022). Description of the linguistic expressions of fractions. *Language Sciences*, 92:101483.
- Bale, A. C. and Barner, D. (2009). The interpretation of functional heads: Using comparatives to explore the mass/count distinction. *Journal of Semantics*, 26(3):217–252.
- Barner, D. and Snedeker, J. (2005). Quantity judgments and individuation: Evidence that mass nouns count. *Cognition*, 97(1):41–66.
- Benbaji-Elhadad, I. and Wehbe, J. (2024). On the semantics of unit fractions. In *Proceedings of Sinn und Bedeutung*, volume 28, pages 131–149.
- Bhatt, R. (2002). The raising analysis of relative clauses: Evidence from adjectival modification. *Natural language semantics*, 10(1):43–90.
- Casati, R. and Varzi, A. C. (1999). *Parts and places: the structures of spatial representation*. MIT press.
- Chomsky, N. (1970). Remarks on nominalization. 1970.
- Chomsky, N. (1981). Lectures on government and binding theory. Dordrecht: Foris.
- Chomsky, N. (1995). *The minimalist program*. MIT press.
- Davies, M. (2010). The corpus of contemporary american english as the first reliable monitor corpus of english. *Literary and linguistic computing*, 25(4):447–464.
- de Hoop, H. (1997). A semantic reanalysis of the partitive constraint. *Lingua*, 103(2-3):151–174.
- de Hoop, H. (2003). Partitivity.
- Deal, A. R. (2015). Interaction and satisfaction in ϕ -agreement. In *Proceedings of NELS*, volume 45, pages 179–192.
- Falco, M. and Zamparelli, R. (2019). Partitives and partitivity. *Glossa: a journal of general linguistics*, 4(1).
- Givón, T. (1981). On the development of the numeral ‘one’ as an indefinite marker. *Folia linguistica historica*, 15(Historica-vol-2-1):35–54.
- Grice, H. P. (1975). Logic and conversation. In *Speech acts*, pages 41–58. Brill.
- Haida, A. and Trinh, T. (2021). Splitting atoms in natural language. *Formal approaches to number in Slavic and beyond*, pages 277–296.
- Hoeksema, J. (1996). Jacob Hoeksema (ed.), Partitives. *Studies on the Syntax and Semantics of Partitive and Related Constructions*, chapter Introduction. Mouton de Gruyter.
- Ionin, T. and Matushansky, O. (2006). The composition of complex cardinals. *Journal of semantics*, 23(4):315–360.
- Ionin, T., Matushansky, O., Ruys, E. G., et al. (2006). Parts of speech: Toward a unified semantics for partitives. In *PROCEEDINGS-NELS*, volume 36, page 357.
- Klockmann, H. (2018). The design of semi-lexicity: Numerals, quantifiers, and kind-expressions.
- Lederer, H. (1977). A german reference grammar.
- Leonard, H. S. and Goodman, N. (1940). The calculus of individuals and its uses¹. *The journal of symbolic logic*, 5(2):45–55.
- Link, G. (1983). The logical analysis of plurals and mass terms: A lattice-theoretical approach. *Formal semantics: The essential readings*, 127:147.
- López Palma, H. (2011). Los numerales partitivos en español. *Moenia*, 17.
- López Palma, H. (2012). Romance fractional numerals. In *22nd Colloquium on Generative Grammar*.
- Moltmann, F. (1997). *Parts and wholes in semantics*. OUP USA.
- Pelletier, F. J. (1975). Non-singular reference: Some preliminaries. *Philosophia*, 5(4):451–465.
- Pesetsky, D. (2013). *Russian case morphology and the syntactic categories*, volume 66. MIT Press.
- Sauerland, U., Anderssen, J., and Yatsushiro, K. (2005). The plural is semantically unmarked. *Linguistic evidence: Empirical, theoretical, and computational perspectives*, pages 413–434.
- Selkirk, E. (1977). Some remarks on noun phrase structure. *Formal Syntax*.
- Simons, P. (1987). *Parts: A study in ontology*. Oxford University Press.
- Sportiche, I. and Charneval, D. (2025). On semantic agreement. In *GLOW*.
- Timberlake, A. (2004). *A reference grammar of Russian*. Cambridge University Press.
- Wagiel, M. (2021). *Subatomic quantification*. Language Science Press.
- Wellwood, A. (2019). *The meaning of more*, volume 12. Oxford University Press.